



RAMCO INSTITUTE OF TECHNOLOGY

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NAAC Accredited



Department of Mechanical Engineering

Ref.: Department Advisory Committee Meeting/2020-21/Odd Semester/3

Date: 21.11.2020

Minutes of the Meeting of Department Advisory Committee Meeting - III

The 3rd Department Advisory Committee Meeting for the AY 2020 – 21 (Odd Semester) was conducted on 21.11.2020 (Saturday) from 03.30 PM to 5:30PM in Google Meet.

Agenda of the meeting:

1. Review of previous Department Advisory Committee Meeting – II
2. Teaching and Learning Process
3. Course Outcomes and Program Outcomes
 - a. Assessment Methods,
 - b. Attainment of Cos, POs, PSOs with program effectiveness
 - c. Action taken for outcome attainment
 - d. Correctiveness and preventiveness
4. Students Performance
5. Department and Industry Interaction
 - a. MoUs
 - b. Industry projects
 - c. Publication with industry
6. Training and Placement Progress with Feedback from Recruiters and Employers
7. Alumni Interaction and Contribution
 - a. Feedback received from the Alumni
8. Faculty contribution
 - a. Participation
 - b. Publication – Conference / Journal / Book Chapter
 - c. Patent & Copy rights
 - d. Research Proposals
9. Activity leads to Quality improvements

03
11.12.20

Members present:

Members present:	Faculty Members – Dept. of Mech. Engg., RIT
1) Mr.N.Rajasekaran – Dy. General Manager, BHEL, Trichirapalli. (Industry Member)	1) Dr.S.Godwin Barnabas, ASCP
2) Dr.L.Srinivasan – Dy. General Manager, VSSC, ISRO (R& D Member)	2) Mr.M.Lakshmanan, AP(Sr.Gr.)
3) Dr.M.Palaninatharaja, Dean P&D TCE – Madurai. (Academic Member)	3) Mr.J.Jerold John Britto, AP(Sr.Gr.)
4) Mr.P.Esakiraja, Asst Manager, Ramco Cements Ltd., Alathithur (Alumni Member)	4) Mr.M.Ashok Kumar, AP(Sr.Gr.)
5) Mr.K.Aravind Kumar, Asst. Manager, Madasamy Nadar Restaurant (Alumni Member)	5) Mr.C.Gururaj, AP(Sr.Gr.)
Internal Members from RIT	6) Mr.M.Sivagaminathan alias Balaji, AP(Sr.Gr.)
1. Dr.S.Rajakarunakaran, Prof & Head / Mech.	7) Mr.G.Prabu ram, AP(Sr.Gr.)
2. Dr.K.Karthikeyan, ASCP/EEE	8) Mr.S.Maharajan, AP
3. Dr.M.Gomathy Nayagam, ASCP/CSE	9) Mr.K.Vigneshwaran, AP
4. Dr.O.Senthilkumar, ASCP/Chemistry	10) Mr.T.Selvasundar, AP
5. Dr.T.Manimaran, ASCP/Maths	11) Mr.S.Valai Ganesh, AP
6. Dr.P.Sureshkumar, ASCP/Mech.	12) Mr.J.Jabinth, AP
7. Dr.V.Sivakumar, ASCP/Mech.	13) Mr.L.Karthikeyan, AP
	14) Mr.M.Santhana Maruthu Pandian, AP
	15) Mr.R.Arun Kumar, AP
	16) Mr.N.L.Sujin, AP
	17) Mr.R.Prabhakaran, AP
	18) Mr.R.Venkatesh, AP
	19) Mr.K.Amudhan, AP
	20) Mr.M.Ramar, AP
	21) Mr.P.Pavithran, AP
	22) Mr.S.Kathiravan, AP

Sl. No.	Agenda	Target Date	Points discussed	Responsibility
1.	Review of previous Department Advisory Committee Meeting – II	Completed	Dr.P.Sureshkumar, ASCP/Mech. presented the minutes of second Department Advisory Committee Meeting. The implementation and suggestions given by the advisory committee members in the previous meet was reviewed by the members.	All the NBA Criteria Coordinators and faculty members
2.	Teaching and Learning Process	Continuous	Innovative practices followed in TLP, Curriculum gap identification, content beyond syllabus, students' feedback were summarized by Mr.R.Arun Kumar, AP/Mech. • The list of innovative practices followed with the methods and ICT tools were described.	All the faculty members & Virtual Lab Coordinator

			"Learning Management Systems – Canvas and Google Classroom are practiced in the department" - The unique delivery methods in the department is divided as "active learning, passive learning, collaborative learning, e-learning and feedback based learning" - Presented the method of transparent evaluation using scheme of evaluation, rubrics, online quizzes and central evaluation. - Added the attainment level of the students have improved marginally using these practices. - Planned to have a tie-up with IUCEE consortium to improve the TLP". - A paper on the impact created through Innovative practices followed is planned to communicate to a journal. - Explained that learning materials of TLP are made available to public for critic reviews in YouTube and Slide share platforms. - University approved value added courses to bridge the curriculum gap was presented and the courses are: Smart Materials & Structures Green Energy Technologies and Management Automation Suite for Smart Systems IoT applications in Mechanical Engineering Surface Coating Technology - Presented how curriculum gap is identified and addressed in individual subjects with two subjects as case studies. - Concluded the presentation with the academic actions taken on feedback collected from the students through Class Committee Meeting, Course Committee Meeting and feedback form. - The virtual laboratory utilization was presented by Mr.J.Jerold John Britto, AP(SG)/Mech. Explained the completed activities and planned activities related to V-labs. "Tie-up with NIT Warangal in developing a virtual lab is under progress". V-Lab utilization details were also presented.	
3.	Course Outcomes, Program Outcomes and Program Specific	Continuous	Mr. J. Jabinth, AP/Mech. presented the overview of the assessment methods. CO-PO attainment for 2016-2020 batch calculation was explained to the members. - Course Exit survey is newly included and calculated for both Theory & Practical from 2019-20 batch with a weightage of 10%. 40% weightage for IAT, 60% weightage for	All the faculty members



	Outcomes		university contributes to 90% of CO attainment along with 10% weightage from CES. - The DAC Members approved the threshold/Target for upcoming semester, which is calculated by taking the maximum value of current, previously set target value in previous academic years. - In first year 4 subjects and in Second year 3 subjects did not attain the target value. In Third Year & Final Year all the subjects attained the target value. - The Alumni Exit survey & Program exit survey were collected via online mode. The collected survey was given 10% weightage each totaling 20% Weightage. - The CO attainment is correlated with mapping to obtain the PO-PSO attainment for each Attributes. The number of subjects mapping the program outcomes were discussed. - The program outcomes & Program Specific Outcomes are determined by taking 80% weightage from Subject attainment and 20% Weightage from PES, Alumni surveys. - The corrective actions taken to improve the attainment were discussed by the presenter.	
4.	Students Performance	Continuous	Mr. G. Prabu ram, AP(Sr.Gr.)/Mech. presented the following points. - Results attained by the students during 2019-2020 (Odd Semester) – 60.89% (232/381) and Even Semester – 100% (381/381) 2020-2021 (Odd Semester (excluding IV Year). - The progresses of internship ➤ 49 students in 2019-20 and 13 students in 2020-21 (Virtual Internship) 9 students have undergone in-plant training - Briefed the student feedback report about Internship/In-plant training, formation of Department Consultative Committee, Rubrics for IPT/Internship review, student presentation and evaluation and suggestions and corrective measures 223 students' participation in Co-Curricular / Extra-Curricular activities (As on date) 41 and 38 projects turned out from final year and third year students, respectively.	All the student members, subject handling faculty members and IPT and Internship In-charge



			- Final year student projects consists of 32 In house projects and 09 Industrial projects - Student achievements during 2019-20 and 2020-21 (as on date) Ref: https://www.ritrjpm.ac.in/images/mechanical/2019-20.pdf https://www.ritrjpm.ac.in/images/mechanical/IPT-2019-20.pdf https://www.ritrjpm.ac.in/departments/mechanical-engg/mechanical-engg-awards-and-achievements.php	
5.	Department and Industry Interaction	Continuous	Mr.K.Amuthan, AP/Mech. Shared the following industry interaction activities - Listed out the number of MoUs with its validity period - Establishment of laboratories through MoUs from RIT – Harita Centre of Excellence and RIT-MEDSBY additive manufacturing centre 13 students placed through MoU companies 8 Industrial problems based projects - Involvement of industrial experts in the project review committee 2 publications with industry person.	All the faculty members &
6.	Training and Placement Progress with Feedback from Recruiters and Employers	Continuous	Mr.M.Lakshmanan, AP(Sr.Gr.)/Mech. explained the following activity during the AY 2019-20. - Aptitude Training, Programming Training, Fundamental of core subject, GD & Mock Interviews, Continuous skill rack practice, quizzes, GATE coaching and class for critical thinking training are given to II, III and IV year students. - Company specific training like TCS NQT, Mock GD, Mock Interview for Kaar Tech, Embed UR, Vuram Tech. 8 Webinars conducted - Opportunity from 31 companies has been given to the students, out of which 7 core companies, 16 IT companies, 1 Knowledge process outsourcing(KPO), 2 sales company, 1 support company and 4 training companies. 73 students placed in 2019-2020 - Collective recruiters (Ramco Systems, Ramco Cements, Harita Techserv, UCAL Fuel Systems, Accurate Steel Forging, TCS, ETHNUS, and Chain System) feedback are collected and the action are taken - Employers' feedback (Ramco Systems, Ramco	All the students, class advisor, mentor and Placement coordinator



			Cements, Harita Techserv, UCAL Fuel Systems and Accurate Steel Forging) and the action taken towards their valuable suggestions. The scheduled future campus drives are (Sundaram Clayton, Hosur, Myunghwa Automotive India – Internship cum Placement, SKF Bearings, JSW Steel, National Instruments (NI) - Certified LabVIEW Associate Developer (Clad).																																	
7.	Alumni Interaction and Contribution	Continuous	Mr.M.Lakshmanan, AP(Sr.Gr.)/Mech. shared the following contribution details of the Alumni over the period. <table><tr><th colspan="4">Alumni Contribution Details</th></tr><tr><th>S.No.</th><th>Date</th><th>Alumni details</th><th>Topic Discussed</th></tr><tr><td>1.</td><td>10.01.20</td><td>Mr.M.Padmanapan, Design Engineer Trainee, Harita Techserv Limited. (2015 – 2019)</td><td>Role of Design Engineer</td></tr><tr><td>2.</td><td>26.02.20</td><td>Mr.Jeevaganesh. C Mr.Arumugam.A Mr.Karthickraj.A Team Leader in RANE, Chennai. (2015 – 2019)</td><td>What Industry Expect from Engineer</td></tr><tr><td>3.</td><td>25.05.20</td><td>Mr.K.Gopinath and Mr.S.Harikrishna (2013 – 2017)</td><td>Gate is a Gateway for new life</td></tr><tr><td>4.</td><td>05.06.20</td><td>Mr.M.Karthick Gowtham , Deputy Registrar (Group I), Cooperative Society. (2013 – 2017)</td><td>Preparation Strategy for TNPSC Exams</td></tr><tr><td>5.</td><td>13.09.20</td><td>K.Ajeethkumar, Managing director of AK Constructions. (2015 – 2019)</td><td>Scope for Entrepreneurs</td></tr><tr><td>6.</td><td>27.09.20</td><td>R.Ajith Kumar, Quality Engineer, CSWIP 3.1 Welding Inspector, ASNT Level II RT, UT, Severn Glocon India Pvt Ltd, Chennai. (2015 – 2019)</td><td>Scope for Jobs in Oil and Petroleum Industry</td></tr></table>	Alumni Contribution Details				S.No.	Date	Alumni details	Topic Discussed	1.	10.01.20	Mr.M.Padmanapan, Design Engineer Trainee, Harita Techserv Limited. (2015 – 2019)	Role of Design Engineer	2.	26.02.20	Mr.Jeevaganesh. C Mr.Arumugam.A Mr.Karthickraj.A Team Leader in RANE, Chennai. (2015 – 2019)	What Industry Expect from Engineer	3.	25.05.20	Mr.K.Gopinath and Mr.S.Harikrishna (2013 – 2017)	Gate is a Gateway for new life	4.	05.06.20	Mr.M.Karthick Gowtham , Deputy Registrar (Group I), Cooperative Society. (2013 – 2017)	Preparation Strategy for TNPSC Exams	5.	13.09.20	K.Ajeethkumar, Managing director of AK Constructions. (2015 – 2019)	Scope for Entrepreneurs	6.	27.09.20	R.Ajith Kumar, Quality Engineer, CSWIP 3.1 Welding Inspector, ASNT Level II RT, UT, Severn Glocon India Pvt Ltd, Chennai. (2015 – 2019)	Scope for Jobs in Oil and Petroleum Industry	Alumni coordinator
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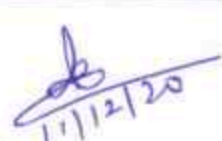
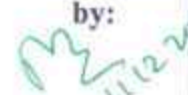
			<table><tr><td>7.</td><td>18.10.20</td><td>Mr.P.Esakkiraja, Assistant Manager – Mechanical, Ramco Cements Ltd, Ariyalur District. (2014 – 2018)</td><td>The Engineer</td></tr><tr><td>8.</td><td>12.11.20</td><td>Mr.P.Ari Ramalingam, Assistant Manager – New Project and Process Development, RANE NSK Steering System Pvt. Ltd. Ahmedabad (2013 – 2017). Mr.B.Vengalabothi, Junior Functional Consultant, Avaniko Technologies Pvt Ltd., Chennai (2013 – 2017).</td><td>College and Industry Environment</td></tr></table> - Delivered the inferences received from the Alumni feedback analysis from 113 Alumnus. - The key feedback from the alumnus about the institute are Good Infrastructure, Food provided in hostel, way of teaching, Online course practice and Rules and regulations followed in the campus	7.	18.10.20	Mr.P.Esakkiraja, Assistant Manager – Mechanical, Ramco Cements Ltd, Ariyalur District. (2014 – 2018)	The Engineer	8.	12.11.20	Mr.P.Ari Ramalingam, Assistant Manager – New Project and Process Development, RANE NSK Steering System Pvt. Ltd. Ahmedabad (2013 – 2017). Mr.B.Vengalabothi, Junior Functional Consultant, Avaniko Technologies Pvt Ltd., Chennai (2013 – 2017).	College and Industry Environment	
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8.	Faculty contribution	Continuous	Dr.V.Sivakumar, ASCP/Mech. addressed the following faculty contribution - Improvement of student faculty ratio (18.72 to 16.04) in the AY 2020-2021 through the recruitment of three faculty members to the department. - Presented the faculty participation in webinar, quiz, FDP, STTP and online courses for the AY 2019-20 and AY 2020-2021 till November. - Highlighted the paper publication (9 Journal papers, 1 article, 17 conference papers, 2 book chapters), 4 Patent filed and 4 Patents Drafting 2 copyrights approved and 1 copyright filed - Status of completed funded projects IE(I) R&D Grant in Aid Scheme-1, - Ongoing projects – DST SSTP & RuTAG IIT Madras, AICTE MODROB – 1,	All the faculty members & RIT RC member								



			- Funds received to conduct program DST – NIMAT – EAC, and IWS, - Proposal submitted in DST, RuTAG, ISHRAE, DST-WMT (4), FAER (7). Further, - Presented the steps taken to improve Research & Development activities of the department. Ref.: Academic Year Wise Faculty Participation & Research and Developmental activities are listed out in the website https://www.ritrjpm.ac.in/departments/mechanical-engg/mechanical-engg-program-organized-program-participation.php https://www.ritrjpm.ac.in/departments/mechanical-engg/mechanical-engg-research-and-development.php	
9.	Activity leads to Quality improvements	Continuous	Mr.J.Jerold John Britto, AP(Sr.Gr.)/Mech. presented the activities that lead to quality improvement <u>Project work</u> - Categorized the projects done by the students as 6 patentable, 3 product development, 12 project completion and 3 eligible to apply for research grant 8 Dept. level prize winners and 3 institute level prize winners in the National Science Day - Ongoing industry base interdisciplinary project through RIT HARITA IOT Home <u>Online course</u> - All the faculty members and students utilized free coupons to do courses in Edx/IIRS - Multi variable calculus course completed and received certificate by all the faculty members - Students involvement in ICTACT – New India Learnton 2020 (10 Courses / 38 students), Anna University approved value added courses (2 credit and 1 credit subjects / 118 students) - AICTE – Universal Human Values course completed by 4 faculty members - RIT IUCEE course – 3 faculty members completed <u>Research</u> - Research Incentive Rs. 36,500/- received by the faculty members - IWCSET – 2020 Participation – 319, Presentation – 46 and publication – 7 - Research talk during the Department Review Meeting	Department of Mechanical Engineering

			Other Programs • ESVC Junior 1.0 for school students (Participation: 36 teams / 360 students and 40 faculty members) • Webinars participation through IITM PALS, TLC of IITM, ICTACT, RIT – IQAC, IRS/ISRO and IUCEE – Entrepreneurship Education Program • 46 Virtual laboratory experiments (MHRD V-Lab) conducted for theory and lab courses • Suggestion received from Parents Teachers meeting on 23.02.20 • Listed out the LMS usage, activities through the GATE class and revised ISO Forms (SV 01, PU 02, PU 04 and AC 11c)	
10.	Members Suggestions for Improvements	Before commencing the next DRM - IV	Mr.N.Rajasekaran – Dy. General Manager, BHEL, Trichirapalli. (Industry Member) • Appreciated the student and faculty contribution by the department. • Suggested to approach the professional societies for funding to conduct program Dr.L. Srinivasan – Dy. General Manager, VSSC, ISRO (R& D Member) • Appreciated the overall presentation by the faculty members and suggested to take necessary action to address the weaker areas represented in the employer's feedback. • Congratulated the winners and participant students in National Science day Project Presentation competition and India skills competition. • Encouraged the faculty members to work in the Additive Manufacturing field and highlighted the opportunities available to work with Indian Space Research Organization. Dr.M.Palaninatharaja, Dean P&D, TCE (Academic member) • Appreciated the content presented in the meeting and suggested to identify the impact analysis and corrective action plan for further improvement to the next academic year. • Advised to enhance fruitful interaction through MoUs with industries and obtain more outcomes. • Advised to analyse CO-PO-PSOs semester wise. Suggested to create a plan for the new academic year and periodically check the progress and its deviation if any.	Department of Mechanical Engineering

		- Recommended to follow the existing procedure adapted in the calculation of CO-PO-PSO and setting the target for the next academic year. Mr.P.Esakkiraja, Asst Manager, Ramco Cements Ltd., Alathithur. (Alumni Member) - Requested to ensure that third and final year students are provided opportunities to refresh their fundamentals in their higher semesters. Mr.K.Aravind Kumar, Asst. Manager, Madasamy Nadar Restaurant, (Alumni Member) - Suggested to create awareness about Entrepreneurship through Institute Innovation Council by conducting program. Dr.S.Rajakarunakaran, Prof. & Head Mech. (Internal Member) - Recommended to disseminate the activities carried out through the industry by the department in the college website. - Motivated the faculty members to do more innovative practices in their subject and laboratory courses and update in the website periodically. Dr.K.Karthikeyan, ASCP/EEE (Internal Member) - Recommended to review periodically, the Internship/In-plant training undergone by the students through Department Consultative Committee. Dr.M.Gomathy Nayagam, ASCP/CSE (Internal Member) - Suggested to create pathway to the interested mechanical engineering students to involve in the Automation and Robotics Club Dr.O.Senthilkumar, ASCP/Chemistry (Internal Member) - Proposed to submit more inter disciplinary project proposals to the governmental and non-governmental agencies.	
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		Dr.T.Manimaran, ASCP/Maths (Internal Member) - Advised to make use of faculty members from mathematics to provide coaching for GATE class attending students through Higher Education Cell and Math Experts Club. Dr.P.Sureshkumar, ASCP/Mech. (Internal Member) - Recommended to provide facilities for the hostellers to come and work in the laboratory after regular working hours. Dr.V.Sivakumar, ASCP/Mech. (Internal Member) - Suggested to constitute a committee and strengthen the quality of projects and get more tangible outcomes.	
11.	Vote of Thanks	Dr.S.Godwin Barnabas, ASCP/Mech. thanked all the internal and external DAC members for attending the meeting.	
Prepared by:  NBA Coordinator		Verified by:  11/12/20 HoD Mechanical	Approved by:  11/12/20 Principal

Copy to:

- 1) Members of DACM
- 2) Faculty members of Mechanical Engineering through e-mail and Canvas Classroom for information and necessary action.
- 3) File